Korean Stroke Registry
Annual Report
2023

Korean Stroke Registry
한국뇌졸중등록사업

KSR Annual Report – **2023**

Korean Stroke Registry

About Korean Stroke Registry (KSR)

Korean Stroke Registry(KSR) is a representative multi-center, prospective, and hospital-based stroke registry in Korea. KSR established to collect data on patients with acute stroke and transient ischemic attack, and to monitor, evaluate, and improve the quality of stroke care in Korea.

In 2001, KSR was launched with 33 major hospitals from 23 medical schools with support from the Korean Stroke Society. In 2006, KSR was progressed with the participation of 66 major hospitals, as one of the six divisions of the Clinical Research Center for Stroke (CRCS) with the support of the Ministry of Health and Welfare, Republic of Korea.

In 2016, the Korean Stroke Society established the KSR steering committee and is operating a new KSR system with the support of the Korea Disease Control and Prevention Agency, Republic of Korea.

As of 2023, 89 hospitals nationwide are participating in the KSR, and 76 hospitals are participating in the KSR Core DB, and participating hospitals are gradually expanding.

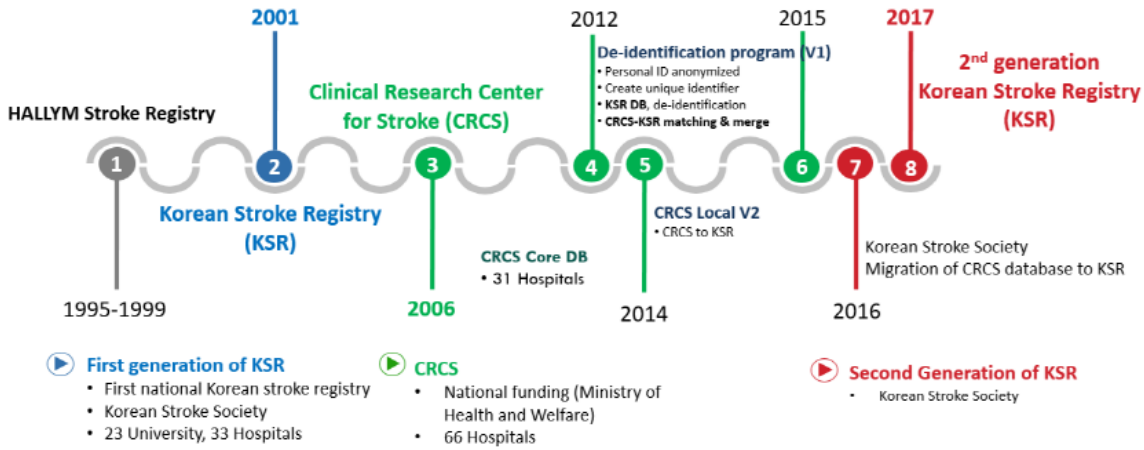

Figure 1. Historical Review of Korean Stroke Registry

Goals of KSR

The overall goal of the KSR is to provide reliable and representative national data on acute stroke care and outcome of patients with all stroke or transient ischemic attack cases in Korea

- To assess the characteristics and secular trend of acute stroke in Korea
- To monitor the quality of acute stroke care
- To implement professional guidelines for acute stroke care
- To monitor the impact of care on post discharge patient outcomes
- To develop a large comprehensive data set for stroke research

Database of KSR

The database of KSR composes a comprehensive and integrated patient management system by supplementing the problems of existing disease-specific and research-specific clinical databases. Reflecting the medical reality, the Core DB was configured to allow input of common data in multiple medical institutions, and the Extended DB was configured to allow data to be added in each study. It is also configured to add long-term prognosis and functional status information (Outcome DB).

In order to manage the data quality of KSR, data are backed up in real-time, and an error check system is automatically applied to reduce errors that may occur during the data entry process. All personal information data has been completely removed through a de-identification program. In addition, the quality of data is improved by managing missing and abnormal input values as a query through the quality management of the database.

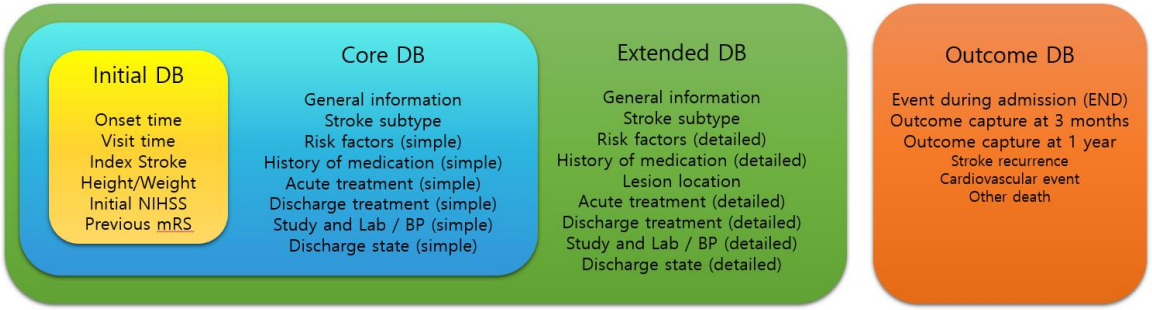

Figure 2. Database structure of Korean Stroke Registry

Participating stroke centers

(KSR Core DB, 76 hospitals, alphabetical order)

The KSR annual report 2023 analyzed the cases of 61 hospitals (marked with an asterisk) that continuously participated in the audit process from Jan 2012 to Dec 2021.

- Asan Asan Medical Center*
- Bucheon Sejong Hospital*
- Bundang Jesaeng Hospital*
- Cha University Bundang Medical Center*
- Changwon Fatima Hospital*
- Changwon Hanmaeum Hospital
- Chonnam National University Hospital*
- Chung-Ang University Hospital*
- Chungbuk National University Hospital*
- Chungnam National University Sejong Hospital*
- Daegu Caholic University Medical Center*
- Daejeon Eulji Medical Center, Eulji University*
- Dong-A University Hospital*
- Dongguk University Ilsan Hospital*
- Dongkang Hospital*
- Ewha Womans University Seoul Hospital
- Gachon University Gil Medical Center*
- Gangnam Severance Hospital*
- Gimpo Woori Hospital

- Gyeonggi Provincial Medical Center Icheon Hospital
- Gyeongsang National University Hospital*
- Gyeongsang National Universtiy Changwon Hospital*
- Hallym University Chuncheon Sacred Heart Hospital*
- Hallym University Dongtan Sacred Heart Hospital*
- Hallym University Sacred Heart Hospital*
- Hana General Hospital*
- Hanyang University Guri Hospital
- Hanyang University Hospital*
- Incheon Paik Hospital
- Inha University Hospital*
- Inje University Busan Paik Hospital*
- Inje University Haeundae Paik Hospital
- Inje University Ilsan Paik Hospital*
- Jeju National University Hospital*
- Jeonbuk National University Hospital*
- Kangbuk Samsung Hospital*
- Kangdong Sacred Heart Hospital*
- Kangwon National University Hospital*
- Keimyung University Dongsan Hospital*
- Konkuk University Medical Center*
- Konyang University Hospital
- Korea University Anam Hospital*
- Korea University Ansan Hospital*
- Korea University Guro Hospital*
- Kosin University Gospel Hospital*
- Kyung Hee University Medical Center*

- Kyungpook National University Chilgok Hospital*
- Kyungpook National University Hospital
- Mokpo Jung-Ang Hospital
- Myongji Hospital*
- Myongji St. Mary's Hospital*
- National Health Insurance Service Ilsan Hospital*
- Nowon Eulji Medical Center, Eulji University*
- Pohang SM Christianity Hospital
- Pohang Stroke and Spine Hospital*
- Pusan National University Hospital*
- Samsung Changwon Hospital
- Samsung Medical Center*
- Seoul Medical Center*
- Seoul National University Bundang Hospital*
- Seoul National University Hospital*
- Severance Hospital*
- Shihwa Medical Center
- SMG-SNU Boramae Medical Center*
- Soonchunhyang University Hospital Bucheon*
- Soonchunhyang University Hospital*
- The Catholic University of Korea Seoul St. Mary's Hospital*
- The Catholic University of Korea St. Vincent's Hospital*
- The Catholic University of Korea, Uijeongbu ST. Mary’s Hospital
- Uijeongbu Eulji Medical Center, Eulji University*
- Ulsan University Hospital*
- University of Ulsan College of Medicine Gangneung Asan Hospital
- Wonju Severance Christian Hospital*

- Yeungnam University Medical Center*
- Yonsei S Hospital*
- Yuseong Sun Hospital*

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Abbreviations

BMI	Body mass index
DBP	Diastolic blood pressure
HDL	High-density lipoprotein
IQR	Interquartile range
LDL	Low-density lipoprotein
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
SBP	Systolic blood pressure
SD	Standard deviation
TG	Triglyceride
TOAST	Trial of ORG 10172 in Acute Stroke Treatment

Global Statistics of KSR

Demographic profile of acute ischemic stroke

- Of 144,412 stroke cases registered in 61 hospitals between Jan 2012 and Dec 2021, number of acute ischemic stroke was 128,607 (89.1%).
- Among the ischemic stroke cases, men were 76,032 (59.1%) and women were 52,575 (40.9%).
- Mean (SD) age of the ischemic stroke cases was 66.1 (12.5) for men and 72.3 (12.8) for women.
- In 2021, among 20,127 ischemic stroke patients, patients aged 85 years or older accounted for 10.9%, and since 2012, in both men and women have been steadily increasing.

Modifiable risk factors of ischemic stroke

- In 2021, Hypertension (67.9%) was the most prevalent risk factor, followed by dyslipidemia (40.1%), diabetes (34.1%), current smoking (21.9%), and atrial fibrillation (19.0%).

Acute treatments of ischemic stroke

- Of the 20,127 ischemic stroke patients registered in 2021, 3,332 (16.6%) received revascularization therapy, of which 1,320 (6.6%) received intravenous thrombolysis, 1,301 (6.5%) intra-arterial thrombolysis, and 711 (3.5%) combined therapy.

Table 1. Number of registered acute stroke patients

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Sum
2012	887 (8.7)	841 (8.3)	928 (9.1)	790 (7.8)	857 (8.4)	884 (8.7)	897 (8.8)	800 (7.9)	815 (8.0)	876 (8.6)	833 (8.2)	746 (7.3)	10154
2013	892 (8.1)	787 (7.2)	944 (8.6)	945 (8.6)	953 (8.7)	867 (7.9)	941 (8.6)	965 (8.8)	886 (8.1)	912 (8.3)	947 (8.6)	952 (8.7)	10991
2014	939 (8.5)	806 (7.3)	892 (8.1)	895 (8.1)	962 (8.7)	867 (7.8)	897 (8.1)	900 (8.1)	941 (8.5)	928 (8.4)	990 (8.9)	1046 (9.5)	11063
2015	929 (7.3)	982 (7.7)	1059 (8.3)	1067 (8.4)	1148 (9.0)	1035 (8.1)	1124 (8.8)	1083 (8.5)	1069 (8.4)	1052 (8.3)	1112 (8.7)	1084 (8.5)	12744
2016	1085 (8.3)	1030 (7.9)	1101 (8.4)	1070 (8.2)	1114 (8.5)	1075 (8.2)	1071 (8.2)	1076 (8.2)	1061 (8.1)	1227 (9.4)	1106 (8.4)	1094 (8.3)	13110
2017	1060 (7.4)	970 (6.8)	1077 (7.6)	1191 (8.4)	1175 (8.2)	1240 (8.7)	1259 (8.8)	1272 (8.9)	1234 (8.7)	1219 (8.6)	1227 (8.6)	1320 (9.3)	14244
2018	1328 (8.5)	1112 (7.1)	1202 (7.7)	1232 (7.9)	1317 (8.4)	1290 (8.3)	1471 (9.4)	1318 (8.4)	1217 (7.8)	1417 (9.1)	1307 (8.4)	1406 (9.0)	15617
2019	1453 (8.4)	1248 (7.2)	1380 (8.0)	1434 (8.3)	1504 (8.7)	1453 (8.4)	1463 (8.4)	1446 (8.3)	1395 (8.0)	1521 (8.8)	1543 (8.9)	1512 (8.7)	17352
2020	1573 (9.3)	1247 (7.4)	1237 (7.4)	1379 (8.2)	1443 (8.6)	1509 (9.0)	1615 (9.6)	1360 (8.1)	1277 (7.6)	1447 (8.6)	1407 (8.4)	1330 (7.9)	16824
2021	1583 (7.1)	1414 (6.3)	1707 (7.7)	1744 (7.8)	1912 (8.6)	2156 (9.7)	2029 (9.1)	1912 (8.6)	1949 (8.7)	1912 (8.6)	2068 (9.3)	1927 (8.6)	22313
Total	11729	10437	11527	11747	12385	12376	12767	12132	11844	12511	12540	12417	144412

Values are presented as number (row percent)

All audited cases registered in KSR

Table 2. Number of acute ischemic stroke patients stratified by sex and age group

Age group	All (n)	Per (%)	Male(n)	Per (%)	Female(n)	Per (%)
<45	5799	4.5	3924	5.2	1875	3.6
45-54	13214	10.3	9712	12.8	3502	6.7
55-64	26016	20.2	19202	25.3	6814	13.0
65-74	34778	27.0	21481	28.3	13297	25.3
75-84	37786	29.4	18105	23.8	19681	37.4
≥85	11014	8.6	3608	4.7	7406	14.1
Mean (SD)	68.6 (13.0)		66.1 (12.5)		72.3 (12.8)	
Total	128607	100	76032	100	52575	100

Based on 128,607 ischemic stroke cases

Table 3. Secular trends of age group proportions in acute ischemic stroke

Age group	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Total										
<45	424 (4.9)	486 (5.1)	518 (5.3)	508 (4.6)	542 (4.8)	601 (4.8)	637 (4.5)	689 (4.3)	560 (3.7)	834 (4.1)
45-54	1003 (11.5)	1146 (11.9)	1095 (11.1)	1182 (10.8)	1207 (10.6)	1273 (10.1)	1438 (10.1)	1557 (9.8)	1455 (9.5)	1858 (9.2)
55-64	1671 (19.2)	1883 (19.6)	1929 (19.6)	2171 (19.8)	2347 (20.6)	2543 (20.2)	2900 (20.4)	3294 (20.7)	3202 (20.9)	4076 (20.3)
65-74	2728 (31.3)	2892 (30.1)	2840 (28.9)	3129 (28.5)	3097 (27.1)	3299 (26.2)	3597 (25.4)	4009 (25.2)	3923 (25.6)	5264 (26.2)
75-84	2338 (26.8)	2605 (27.1)	2728 (27.8)	3238 (29.5)	3374 (29.6)	3850 (30.6)	4346 (30.6)	4824 (30.4)	4575 (29.9)	5908 (29.4)
≥85	544 (6.2)	596 (6.2)	711 (7.2)	753 (6.9)	842 (7.4)	1014 (8.1)	1269 (8.9)	1515 (9.5)	1583 (10.3)	2187 (10.9)
Men										
<45	292 (5.7)	331 (6.0)	356 (6.2)	327 (5.2)	365 (5.5)	426 (5.6)	431 (5.2)	475 (5.0)	354 (3.8)	567 (4.7)
45-54	761 (14.9)	853 (15.4)	781 (13.7)	881 (13.9)	896 (13.5)	950 (12.6)	1043 (12.6)	1135 (11.9)	1060 (11.5)	1352 (11.1)
55-64	1204 (23.6)	1347 (24.3)	1379 (24.1)	1551 (24.5)	1717 (25.8)	1884 (25.0)	2137 (25.9)	2478 (26.0)	2404 (26.1)	3101 (25.5)
65-74	1619 (31.7)	1664 (30.0)	1675 (29.3)	1847 (29.2)	1836 (27.6)	2058 (27.3)	2205 (26.7)	2542 (26.7)	2585 (28.1)	3450 (28.4)
75-84	1041 (20.4)	1156 (20.9)	1288 (22.5)	1487 (23.5)	1585 (23.8)	1896 (25.1)	2056 (24.9)	2388 (25.1)	2282 (24.8)	2926 (24.1)
≥85	183 (3.6)	193 (3.5)	241 (4.2)	238 (3.8)	260 (3.9)	331 (4.4)	387 (4.7)	495 (5.2)	530 (5.8)	750 (6.2)
Women										
<45	132 (3.7)	155 (3.8)	162 (4.0)	181 (3.9)	177 (3.7)	175 (3.5)	206 (3.5)	214 (3.4)	206 (3.4)	267 (3.3)
45-54	242 (6.7)	293 (7.2)	314 (7.7)	301 (6.5)	311 (6.5)	323 (6.4)	395 (6.7)	422 (6.6)	395 (6.5)	506 (6.3)
55-64	467 (12.9)	536 (13.2)	550 (13.4)	620 (13.3)	630 (13.3)	659 (13.1)	763 (12.9)	816 (12.8)	798 (13.1)	975 (12.2)
65-74	1109 (30.7)	1228 (30.2)	1165 (28.4)	1282 (27.6)	1261 (26.5)	1241 (24.6)	1392 (23.5)	1467 (23.0)	1338 (22.0)	1814 (22.7)
75-84	1297 (35.9)	1449 (35.7)	1440 (35.1)	1751 (37.7)	1789 (37.7)	1954 (38.8)	2290 (38.6)	2436 (38.2)	2293 (37.7)	2982 (37.4)
≥85	361 (10.0)	403 (9.9)	470 (11.5)	515 (11.1)	582 (12.3)	683 (13.6)	882 (14.9)	1020 (16.0)	1053 (17.3)	1437 (18.0)

Based on 128,607 ischemic stroke cases

Values are presented as number (column percent)

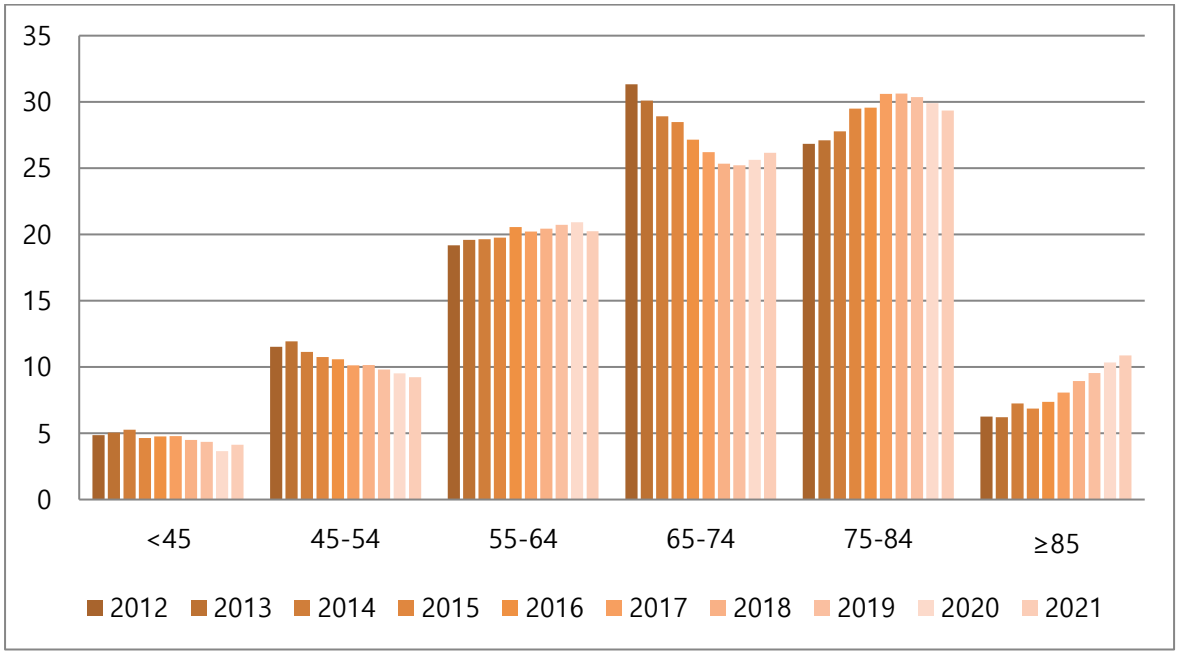
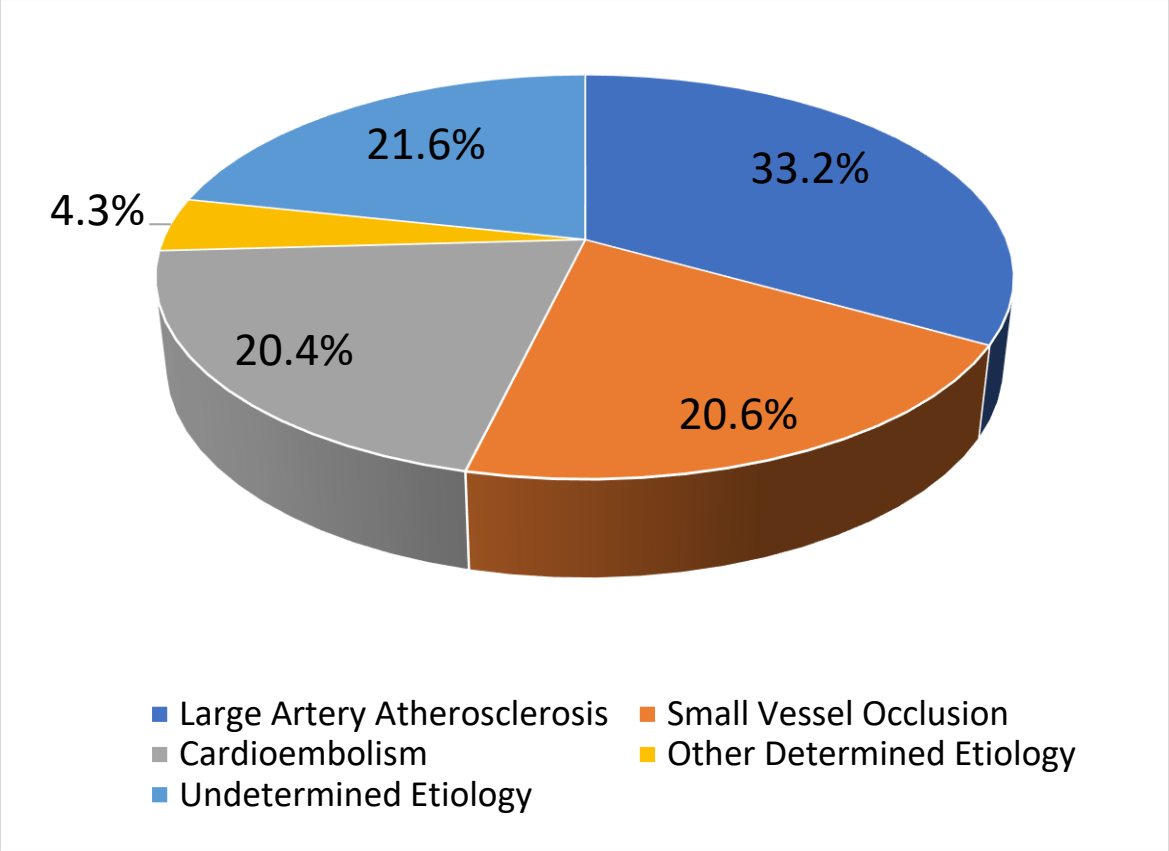


Figure 3. Secular trends of age group proportions in acute ischemic stroke

Table 4. Frequency of TOAST classification subtypes

	Frequency (n)	Percentage (%)
Large Artery Atherosclerosis	42329	32.9
Small Vessel Occlusion	26219	20.4
Cardioembolism	25986	20.2
Other Determined Etiology	5487	4.3
Undetermined Etiology - 2 or more	6690	5.2
Undetermined Etiology - Negative	12384	9.6
Undetermined Etiology - Incomplete work-ups	8475	6.6
Missing	1039	0.8

Based on 128,609 ischemic stroke cases



Based on 127,520 ischemic stroke cases

Figure 4. TOAST classification

Table 5. Secular trends of TOAST Classification

	Number of patients (%)									
	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Large Artery Atherosclerosis	3149 (37.1)	3190 (34.4)	3240 (33.4)	3522 (32.2)	3749 (33.0)	4084 (32.7)	4569 (32.3)	5004 (31.6)	5104 (33.4)	6718 (33.5)
Small Vessel Occlusion	1606 (18.9)	1892 (20.4)	1821 (18.8)	2158 (19.7)	2276 (20.0)	2606 (20.9)	2953 (20.9)	3319 (20.9)	3111 (20.4)	4477 (22.3)
Cardioembolism	1815 (21.4)	1976 (21.3)	2162 (22.3)	2370 (21.7)	2307 (20.3)	2467 (19.8)	2878 (20.3)	3201 (20.2)	2972 (19.5)	3838 (19.1)
Other Determined Etiology	236 (2.8)	349 (3.8)	361 (3.7)	370 (3.4)	412 (3.6)	512 (4.1)	781 (5.5)	776 (4.9)	763 (5.0)	927 (4.6)
Undetermined Etiology - 2 or more	428 (5.0)	422 (4.5)	484 (5.0)	609 (5.6)	595 (5.2)	663 (5.3)	684 (4.8)	798 (5.0)	887 (5.8)	1120 (5.6)
Undetermined Etiology - Negative	562 (6.6)	676 (7.3)	765 (7.9)	1052 (9.6)	1140 (10.0)	1345 (10.8)	1391 (9.8)	1675 (10.6)	1624 (10.6)	2154 (10.7)
Undetermined Etiology - Incomplete work-ups	696 (8.2)	771 (8.3)	861 (8.9)	850 (7.8)	882 (7.8)	807 (6.5)	901 (6.4)	1080 (6.8)	811 (5.3)	816 (4.1)
Total	8492	9276	9694	10931	11361	12484	14157	15853	15272	20050

Based on 127,570 ischemic stroke cases

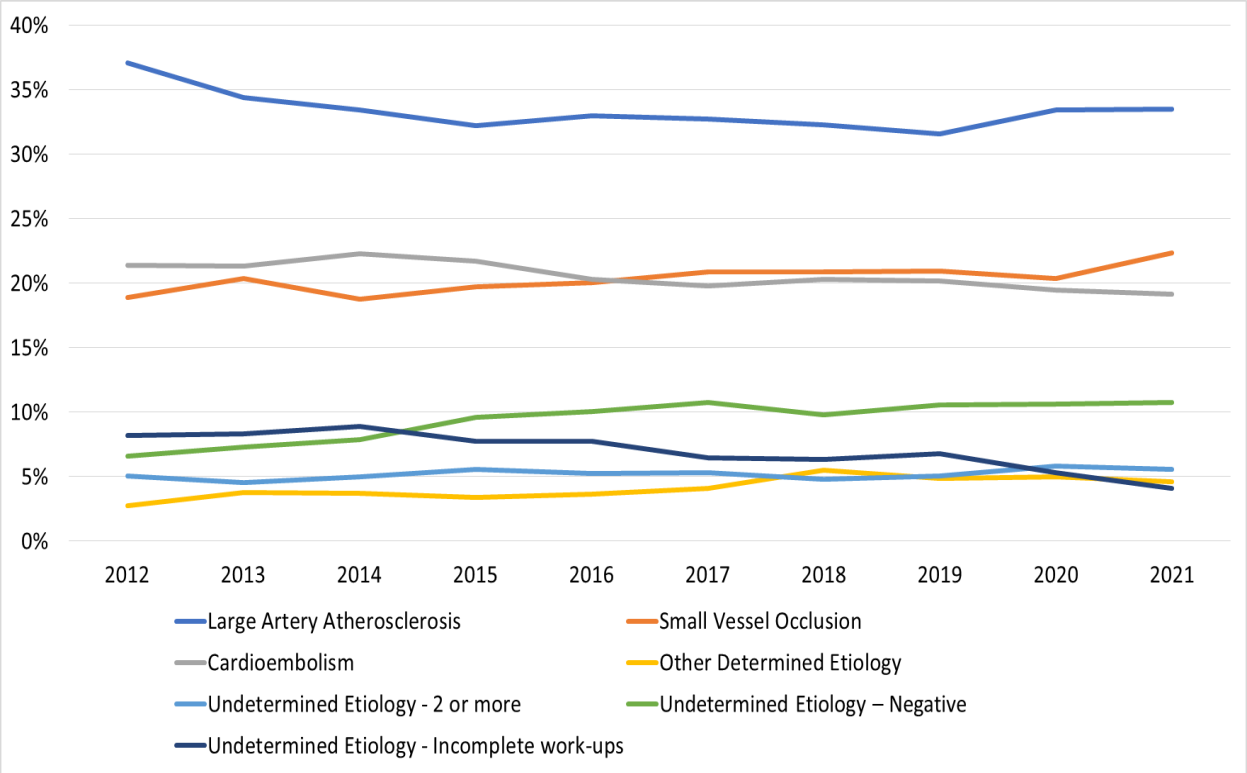


Figure 5. Secular trends of proportions in TOAST Classification

Table 6. Time interval between symptom onset and hospital arrival

	Number of prescription (%)										
	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Jeonbuk	Gwangju & Jeonnam	Chungbuk	Jeju	Overall
≤3 hour	8261 (22.1)	8283 (24.4)	7228 (27.6)	1460 (28.2)	4431 (23.3)	906 (24.9)	127 (19.7)	2889 (27.1)	1228 (28.9)	917 (28.5)	35730 (24.8)
3-6 hour	3384 (9.1)	3404 (10.0)	3394 (13.0)	516 (10.0)	2119 (11.1)	418 (11.5)	65 (10.1)	1786 (16.8)	445 (10.5)	374 (11.6)	15905 (11.0)
6-12 hour	4956 (13.3)	4449 (13.1)	3875 (14.8)	713 (13.8)	2501 (13.2)	475 (13.1)	86 (13.3)	1629 (15.3)	604 (14.2)	473 (14.7)	19761 (13.7)
12-24 hour	6493 (17.4)	5955 (17.5)	4471 (17.1)	879 (17.0)	3518 (18.5)	657 (18.1)	123 (19.1)	1581 (14.8)	749 (17.6)	550 (17.1)	24976 (17.3)
24-36 hour	2941 (7.9)	2429 (7.2)	1752 (6.7)	414 (8.0)	1394 (7.3)	255 (7.0)	58 (9.0)	901 (8.5)	270 (6.3)	224 (7.0)	10638 (7.4)
36-48 hour	2150 (5.8)	1722 (5.1)	1047 (4.0)	260 (5.0)	1020 (5.4)	195 (5.4)	26 (4.0)	300 (2.8)	233 (5.5)	124 (3.8)	7077 (4.9)
2-7 day	7708 (20.6)	6402 (18.9)	3734 (14.3)	835 (16.1)	3398 (17.9)	636 (17.5)	113 (17.5)	1302 (12.2)	666 (15.6)	520 (16.1)	25314 (17.6)
>7 day	1446 (3.9)	1308 (3.9)	676 (2.6)	94 (1.8)	635 (3.3)	93 (2.6)	47 (7.3)	265 (2.5)	61 (1.4)	41 (1.3)	4666 (3.2)
TOTAL	37339	33952	26177	5171	19016	3635	645	10653	4256	3223	144067

Based on 144,067 acute stroke patients

Table 7. Secular trends of categorized symptom onset to arrival time

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
median hr (IQR)	12 (3 - 39)	12 (3 - 39)	12 (3 - 37)	12 (3 - 40)	11 (3 - 38)	12 (3 - 36)	12 (3 - 36)	12 (3 - 37)	12 (3 - 35)	13 (3 - 40)
≤3 hour	2404 (23.8)	2659 (24.4)	2833 (25.7)	3155 (24.8)	3362 (25.7)	3560 (25.0)	3929 (25.2)	4425 (25.5)	4139 (24.6)	5264 (23.6)
3-6 hour	1231 (12.2)	1274 (11.7)	1216 (11.0)	1380 (10.9)	1462 (11.2)	1581 (11.1)	1767 (11.3)	1867 (10.8)	1837 (10.9)	2290 (10.3)
6-12 hour	1364 (13.5)	1513 (13.9)	1459 (13.3)	1717 (13.5)	1839 (14.1)	1975 (13.9)	2082 (13.3)	2389 (13.8)	2382 (14.2)	3041 (13.6)
12-24 hour	1696 (16.8)	1793 (16.5)	1883 (17.1)	2070 (16.3)	2179 (16.7)	2449 (17.2)	2778 (17.8)	3050 (17.6)	3038 (18.1)	4040 (18.1)
24-36 hour	741 (7.3)	781 (7.2)	810 (7.4)	975 (7.7)	869 (6.7)	1088 (7.6)	1174 (7.5)	1234 (7.1)	1284 (7.6)	1682 (7.5)
36-48 hour	507 (5.0)	509 (4.7)	535 (4.9)	690 (5.4)	625 (4.8)	717 (5.0)	750 (4.8)	825 (4.8)	777 (4.6)	1142 (5.1)
2-7 day	1839 (18.2)	1998 (18.3)	1894 (17.2)	2271 (17.9)	2273 (17.4)	2394 (16.8)	2690 (17.2)	3061 (17.6)	2842 (16.9)	4052 (18.2)
>7 day	317 (3.1)	362 (3.3)	380 (3.5)	441 (3.5)	457 (3.5)	462 (3.2)	444 (2.8)	493 (2.8)	517 (3.1)	793 (3.6)

Based on 144,067 acute stroke patients

Table 7-1. Secular trends of categorized symptom onset (last normal time) to arrival time

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
median hr (IQR)	12 (3 - 39)	12 (3 - 39)	12 (3 - 37)	12 (3 - 40)	11 (3 - 38)	12 (3 - 36)	12 (3 - 36)	12 (3 - 37)	12 (3 - 35)	13 (3 - 40)
≤2 hour	1787 (17.7)	2007 (18.4)	2104 (19.1)	2388 (18.8)	2610 (20.0)	2667 (18.7)	2995 (19.2)	3436 (19.8)	3180 (18.9)	4083 (18.3)
≤3.5 hour	2670 (26.4)	2919 (26.8)	3065 (27.8)	3480 (27.4)	3675 (28.1)	3865 (27.2)	4323 (27.7)	4817 (27.8)	4541 (27.0)	5753 (25.8)
≤24 hour	6695 (66.3)	7239 (66.5)	7391 (67.1)	8320 (65.5)	8842 (67.7)	9565 (67.2)	10556 (67.6)	11731 (67.6)	11396 (67.8)	14635 (65.6)

Based on 144,067 acute stroke patients

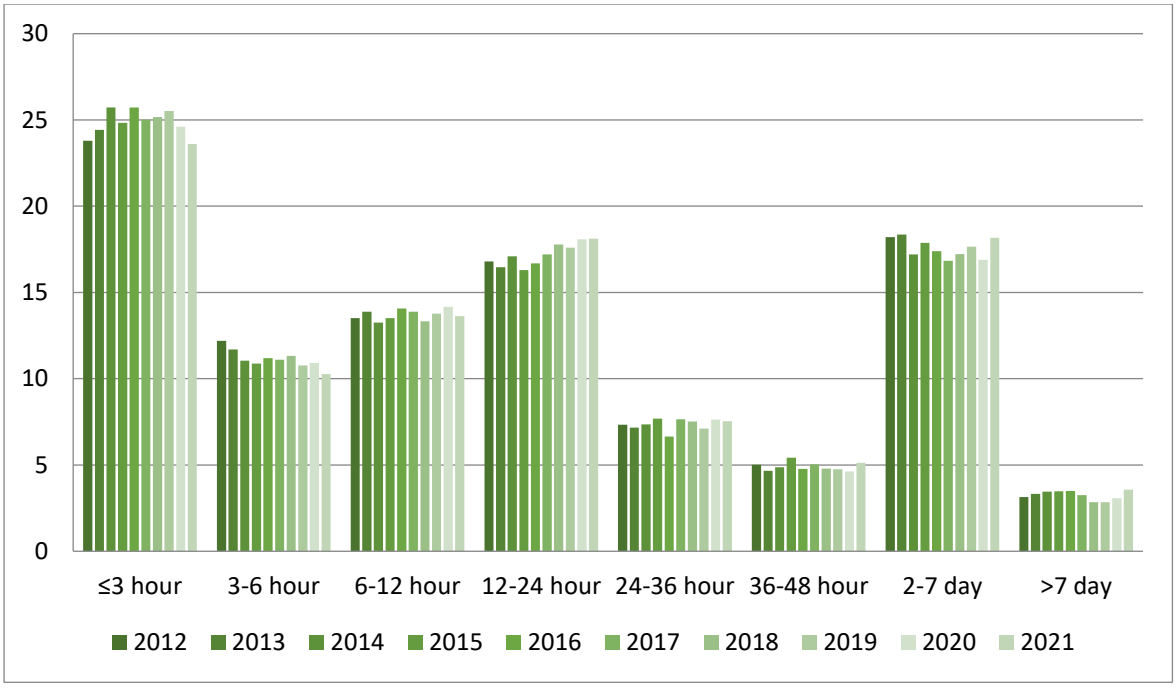


Figure 6. Secular trends of categorized symptom onset to arrival time

Table 8. Secular trends of anthropometric measurements of acute ischemic stroke and transient ischemic attack patients

	Overall, mean (SD)				
	2012-2013	2014-2015	2016-2017	2018-2019	2020-2021
Height (cm)	162.2 (9.0)	162.2 (9.0)	162.5 (9.0)	162.6 (9.0)	162.8 (9.1)
Weight (kg)	62.5 (11.5)	62.6 (11.8)	63.2 (12.1)	63.3 (12.2)	63.8 (12.3)
BMI (kg/m^2)	23.7 (3.4)	23.7 (3.5)	23.8 (3.5)	23.8 (3.6)	24.0 (3.6)
Initial SBP (mmHg)	146.5 (27.1)	146.2 (27.1)	148.8 (27.7)	151.3 (28.4)	153.6 (28.5)
Initial DBP (mmHg)	84.8 (15.2)	84.5 (15.4)	84.6 (15.9)	85.1 (16.5)	86.3 (16.9)
	Men, mean (SD)				
Height (cm)	167.9 (6.1)	167.7 (6.4)	168.0 (6.3)	168.0 (6.3)	168.1 (6.4)
Weight (kg)	67.1 (10.4)	67.3 (10.6)	67.9 (10.9)	68.0 (11.1)	68.5 (11.2)
BMI (kg/m^2)	23.8 (3.2)	23.9 (3.2)	24.0 (3.2)	24.0 (3.3)	24.2 (3.4)
Initial SBP (mmHg)	145.9 (26.4)	146.2 (26.5)	148.7 (27.1)	150.9 (27.8)	153.2 (27.8)
Initial DBP (mmHg)	85.2 (14.9)	85.0 (15.2)	85.3 (15.8)	85.8 (16.5)	87.0 (16.8)
	Women, mean (SD)				
Height (cm)	154.4 (6.0)	154.7 (6.2)	154.7 (6.1)	154.8 (6.1)	154.7 (6.2)
Weight (kg)	56.0 (9.8)	56.2 (10.1)	56.5 (10.4)	56.4 (10.3)	56.7 (10.3)
BMI (kg/m^2)	23.5 (3.7)	23.4 (3.8)	23.5 (3.8)	23.5 (3.9)	23.6 (3.9)
Initial SBP (mmHg)	147.2 (27.9)	146.3 (28.0)	148.9 (28.6)	151.9 (29.2)	154.3 (29.3)
Initial DBP (mmHg)	84.2 (15.5)	83.9 (15.6)	83.5 (16.0)	84.0 (16.4)	85.3 (16.9)

Based on 137,846 acute ischemic stroke and transient ischemic attack patients

Note: The outlier cases are excluded from the analysis in the following cases: height, not between 100-250;

weight, not between 20-200; initial systolic BP, not between 50-300; initial diastolic BP, more than 250.
BMI: body mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure.

Table 9. Secular trends of prestroke mRS score

Prestroke mRS score	Number (%)					
	Total	2012-2013	2014-2015	2016-2017	2018-2019	2020-2021
0	83590 (77.1)	14234 (77.7)	14712 (70.7)	19163 (79.9)	23666 (78.7)	26347 (74.4)
1	9279 (8.6)	1332 (7.3)	2083 (10.0)	1768 (7.4)	2614 (8.7)	3473 (9.8)
2	6082 (5.6)	911 (5.0)	1709 (8.2)	1258 (5.2)	1396 (4.6)	2017 (5.7)
3	5373 (5.0)	954 (5.2)	1249 (6.0)	1046 (4.4)	1425 (4.7)	2017 (5.7)
4	2812 (2.6)	564 (3.1)	660 (3.2)	549 (2.3)	692 (2.3)	1143 (3.2)
5	1341 (1.2)	319 (1.7)	388 (1.9)	205 (0.9)	282 (0.9)	430 (1.2)

Based on 128,606 ischemic stroke cases

Table 10. Secular trends of history of major vascular risk factors

Risk factors and prior managements	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Number of patients	128609	8708	9608	9821	10981	11409	12580	14187	15888	15298	20129
Hypertension	85383 (66.4)	5927 (68.1)	6385 (66.5)	6507 (66.3)	6911 (62.9)	7389 (64.8)	8170 (64.9)	9438 (66.5)	10614 (66.8)	10383 (67.9)	13659 (67.9)
History of Hypertension	76523 (90.8)	5270 (88.9)	5789 (90.7)	6005 (92.3)	6445 (93.3)	6806 (92.1)	7324 (91.8)	8151 (90.6)	9489 (90.9)	9184 (89.5)	12060 (89.4)
Diagnosed at the index admission	7768 (9.2)	656 (11.1)	596 (9.3)	502 (7.7)	466 (6.7)	583 (7.9)	655 (8.2)	844 (9.4)	950 (9.1)	1081 (10.5)	1435 (10.6)
Diabetes	42487 (33.0)	2910 (33.4)	3183 (33.1)	3196 (32.5)	3437 (31.3)	3652 (32.0)	4070 (32.4)	4671 (32.9)	5263 (33.1)	5241 (34.3)	6864 (34.1)
History of Diabetes	36542 (87.1)	2529 (86.9)	2726 (85.6)	2843 (89.0)	3125 (90.9)	3250 (89.0)	3468 (87.1)	3825 (85.9)	4516 (87.0)	4405 (85.0)	5855 (86.5)
Diagnosed at the index admission	5411 (12.9)	380 (13.1)	457 (14.4)	353 (11.0)	312 (9.1)	401 (11.0)	514 (12.9)	627 (14.1)	675 (13.0)	775 (15.0)	917 (13.5)
Dyslipidemia	43854 (34.1)	2787 (32.0)	2770 (28.8)	2818 (28.7)	2945 (26.8)	3493 (30.6)	3791 (30.1)	4811 (33.9)	6176 (38.9)	6183 (40.4)	8080 (40.1)
History of Dyslipidemia	27459 (63.9)	1462 (52.5)	1648 (59.5)	1748 (62.0)	1968 (66.8)	2399 (68.7)	2516 (68.7)	2869 (64.8)	3742 (62.3)	3849 (63.1)	5258 (66.2)
Diagnosed at the index admission	15492 (36.1)	1324 (47.5)	1122 (40.5)	1070 (38.0)	977 (33.2)	1094 (31.3)	1145 (31.3)	1558 (35.2)	2267 (37.7)	2247 (36.9)	2688 (33.8)
Current smoker	45545 (35.4)	2221 (25.7)	2425 (25.4)	2339 (23.9)	2267 (21.0)	2521 (22.4)	2867 (22.8)	3106 (21.9)	3646 (23.0)	3574 (23.4)	6289 (31.2)
Atrial fibrillation	29378 (22.9)	1811 (20.8)	1887 (19.6)	2089 (21.3)	2318 (21.1)	2269 (19.9)	2411 (19.2)	2899 (20.4)	3199 (20.1)	3102 (20.3)	4412 (21.9)
History of Atrial fibrillation	10794 (8.4)	1074 (59.3)	1074 (56.9)	1132 (54.2)	1397 (60.3)	1319 (58.4)	1277 (54.4)	1491 (53.5)	1614 (51.1)	1529 (49.8)	1395 (6.9)
Diagnosed at the index admission	4856 (3.8)	737 (40.7)	812 (43.1)	957 (45.8)	919 (39.7)	940 (41.6)	1072 (45.6)	1298 (46.5)	1542 (48.9)	1542 (50.2)	477 (2.4)

Based on 128,609 ischemic stroke cases. Percentages of management categories were calculated based on the frequency of each risk factor.

Table 10-1. Regional variation of vascular risk factors and their management before index stroke

Risk factors and prior managements	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Jeonbuk	Gwangju & Jeonnam	Chungbuk	Jeju
Number of patients	34172	30474	22269	4609	16379	3339	600	10327	3430	3010
Hypertension	23468 (68.7)	20910 (68.6)	14566 (65.4)	3130 (67.9)	10818 (66.0)	2094 (62.7)	392 (65.3)	5911 (57.2)	2126 (62.0)	1968 (65.4)
History of Hypertension	20423 (90.9)	18804 (89.9)	12523 (86.2)	2893 (92.5)	9778 (90.8)	2022 (96.6)	384 (98.0)	5805 (98.2)	2047 (96.3)	1844 (93.7)
Diagnosed at the index admission	2040 (9.1)	2103 (10.1)	2009 (13.8)	235 (7.5)	992 (9.2)	72 (3.4)	8 (2.0)	106 (1.8)	79 (3.7)	124 (6.3)
Diabetes	11975 (35.0)	10450 (34.3)	7117 (32.0)	1470 (31.9)	5250 (32.1)	1112 (33.3)	196 (32.7)	2910 (28.2)	1108 (32.3)	899 (29.9)
History of Diabetes	10002 (87.0)	8791 (84.1)	5994 (84.4)	1298 (88.5)	4667 (89.4)	1010 (90.8)	185 (94.4)	2806 (96.4)	1059 (95.6)	730 (81.2)
Diagnosed at the index admission	1488 (13.0)	1659 (15.9)	1104 (15.6)	169 (11.5)	556 (10.6)	102 (9.2)	11 (5.6)	104 (3.6)	49 (4.4)	169 (18.8)
Dyslipidemia	14329 (41.9)	11538 (37.9)	7099 (31.9)	933 (20.2)	6202 (37.9)	646 (19.3)	109 (18.2)	1572 (15.2)	476 (13.9)	950 (31.6)
History of Dyslipidemia	8360 (62.0)	8307 (72.0)	3573 (50.4)	836 (89.7)	3664 (59.6)	527 (81.6)	101 (92.7)	1018 (64.8)	385 (80.9)	688 (72.4)
Diagnosed at the index admission	5130 (38.0)	3230 (28.0)	3514 (49.6)	96 (10.3)	2488 (40.4)	119 (18.4)	8 (7.3)	554 (35.2)	91 (19.1)	262 (27.6)
Current smoker	12502 (36.6)	11475 (37.7)	8994 (40.4)	1489 (32.3)	4982 (30.4)	821 (24.6)	178 (29.7)	3080 (29.8)	1120 (32.7)	904 (30.0)
Atrial fibrillation	7731 (22.7)	7624 (25.0)	5638 (25.4)	1012 (22.0)	3378 (20.9)	520 (16.0)	102 (17.0)	1938 (18.8)	712 (20.8)	723 (24.0)
History of Atrial fibrillation	3529 (10.3)	2595 (8.5)	2305 (10.4)	338 (7.3)	882 (5.5)	117 (3.6)	39 (6.5)	582 (5.6)	296 (8.6)	111 (3.7)
Diagnosed at the index admission	1194 (3.5)	1224 (4.0)	962 (4.3)	135 (2.9)	470 (2.9)	101 (3.1)	36 (6.0)	555 (5.4)	109 (3.2)	70 (2.3)

Table 11. NIHSS score at hospital arrival

Region	Mean (SD)	Median (IQR)	Min-Max
Seoul	5.1 (5.6)	3 (1 - 6)	0 - 41
Gyeonggi & & Incheon	5.6 (6.2)	3 (1 - 7)	0 - 42
Gyeongnam & Busan	5.9 (6.1)	4 (2 - 8)	0 - 40
Chungnam & Daejeon	5.2 (6.0)	3 (1 - 7)	0 - 38
Gyeongbuk & Daegu	5.3 (6.1)	3 (1 - 7)	0 - 42
Gangwon	5.0 (5.6)	3 (1 - 6)	0 - 37
Jeonbuk	4.5 (4.4)	3 (1 - 6)	0 - 29
Gwangju & Jeonnam	5.1 (5.5)	3 (1 - 8)	0 - 31
Chungbuk	5.4 (5.7)	3 (1 - 7)	0 - 35
Jeju	6.8 (7.0)	4 (2 - 10)	0 - 38
Total	5.4 (5.9)	3 (1 - 7)	0 - 42

Based on 128,606 ischemic stroke cases. NIHSS: National Institutes of Health Stroke Scale

Table 11-1. Secular trends of proportion in minor (NIHSS≤3) and severe stroke (NIHSS>15)

Year of admission	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
NIHSS, median (IQR)	3 (1 - 7)	4 (1 - 8)	3 (1 - 7)	4 (1 - 8)	3 (1 - 7)	3 (1 - 8)	3 (1 - 7)	3 (1 - 7)	3 (1 - 7)	3 (1 - 7)	3 (1 - 6)
NIHSS <4, %	66606 (51.8)	4306 (49.4)	4883 (50.8)	4823 (49.1)	5738 (52.3)	5842 (51.2)	6449 (51.3)	7174 (50.6)	8408 (52.9)	7964 (52.1)	11019 (54.7)
NIHSS ≥15, %	13548 (10.5)	973 (11.2)	1094 (11.4)	1210 (12.3)	1232 (11.2)	1246 (10.9)	1307 (10.4)	1529 (10.8)	1628 (10.2)	1545 (10.1)	1784 (8.9)

Table 12. Profile of method of revascularization therapies for ischemic strokes

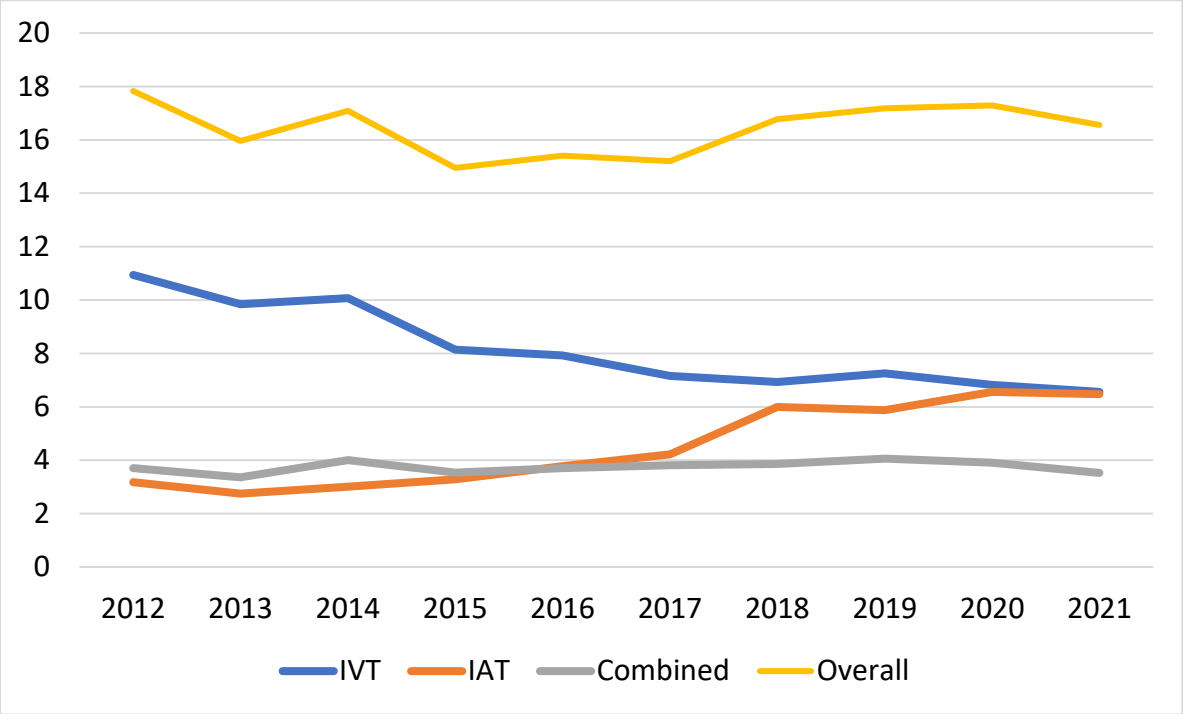
Region	IVT	IAT	Combined	Total
Seoul	2147 (6.6)	1072 (3.3)	873 (2.7)	4092 (12.5)
Gyeonggi & & Incheon	2001 (6.7)	1551 (5.2)	1155 (3.9)	4707 (15.7)
Gyeongnam & Busan	2171 (10.4)	1379 (6.6)	1087 (5.2)	4637 (22.3)
Chungnam & Daejeon	421 (9.1)	96 (2.1)	123 (2.7)	640 (13.9)
Gyeongbuk & Daegu	1005 (6.3)	964 (6.0)	546 (3.4)	2515 (15.7)
Gangwon	268 (9.2)	83 (2.9)	105 (3.6)	456 (15.7)
Jeonbuk	93 (15.5)	42 (7.0)	25 (4.2)	160 (26.7)
Gwangju & Jeonnam	1028 (10.0)	534 (5.2)	414 (4.0)	1976 (19.1)
Chungbuk	203 (5.9)	201 (5.9)	190 (5.6)	594 (17.4)
Jeju	304 (10.1)	191 (6.3)	156 (5.2)	651 (21.6)
Total	9641 (7.8)	6113 (4.9)	4674 (3.8)	20428 (16.4)

Based on 124,247 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Table 13. Secular trends of revascularization modalities

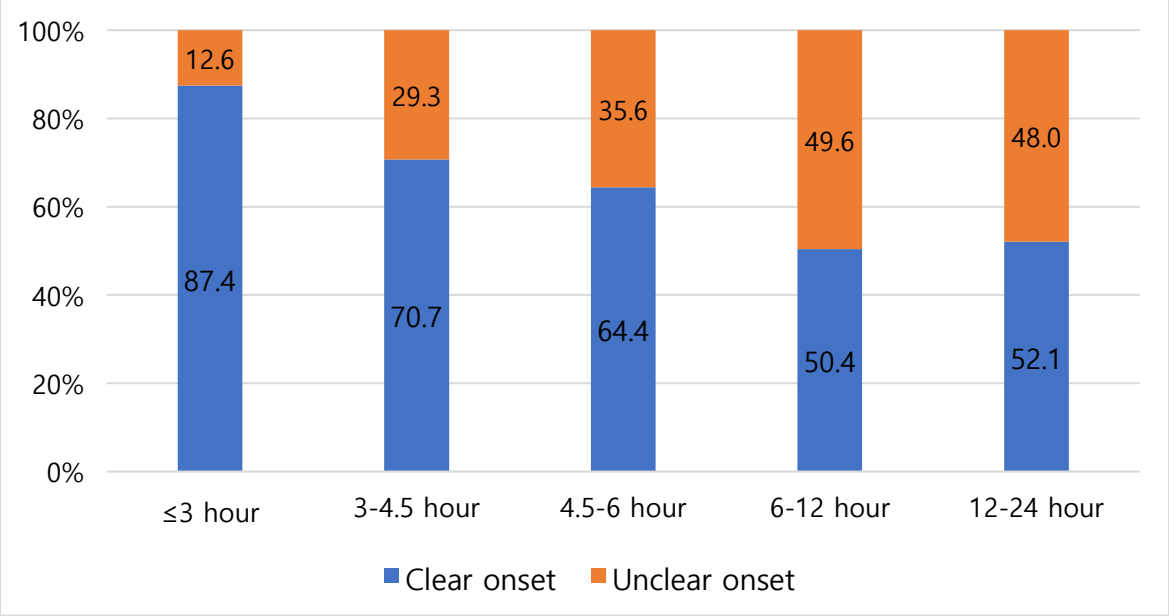
	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
IVT	730 (10.9)	815 (9.8)	932 (10.1)	876 (8.1)	894 (7.9)	896 (7.2)	983 (6.9)	1152 (7.3)	1043 (6.8)	1320 (6.6)	9641 (7.8)
IAT	212 (3.2)	228 (2.8)	279 (3.0)	353 (3.3)	427 (3.8)	528 (4.2)	850 (6.0)	932 (5.9)	1003 (6.6)	1301 (6.5)	6113 (4.9)
Combined	247 (3.7)	278 (3.4)	371 (4.0)	381 (3.5)	419 (3.7)	477 (3.8)	547 (3.9)	645 (4.1)	598 (3.9)	711 (3.5)	4674 (3.8)
Overall	1189 (17.8)	1321 (16.0)	1582 (17.1)	1610 (15.0)	1740 (15.4)	1901 (15.2)	2380 (16.8)	2729 (17.2)	2644 (17.3)	3332 (16.6)	20428 (16.4)

Based on 124,247 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis



Based on 124,247 ischemic stroke cases. IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Figure 7. Secular trends of revascularization modalities



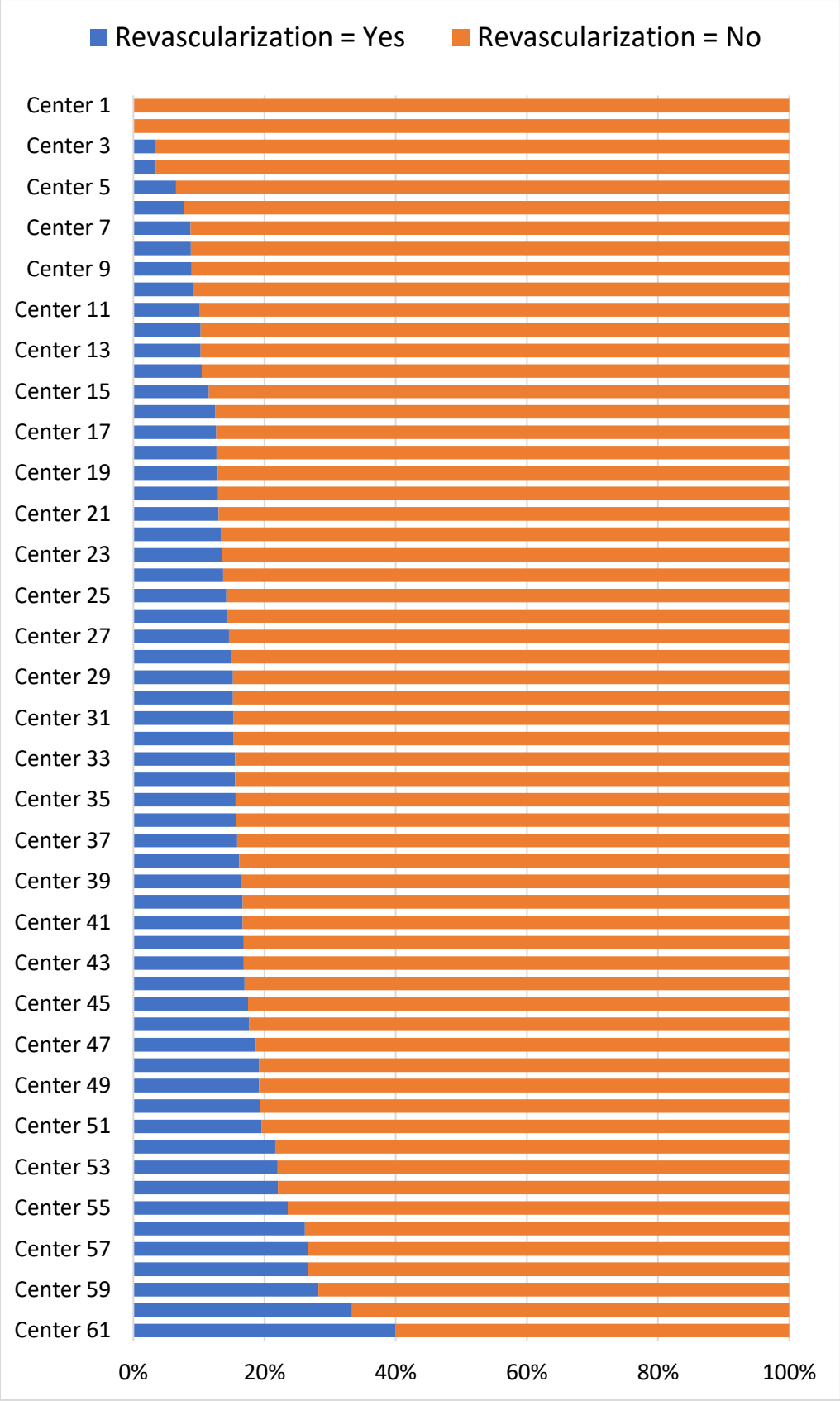
Based on 84,139 ischemic stroke cases

Figure 8. Proportions of clear onset, unclear onset by categorized onset (LNT) to arrival time

Table 14. Secular trends of laboratory tests

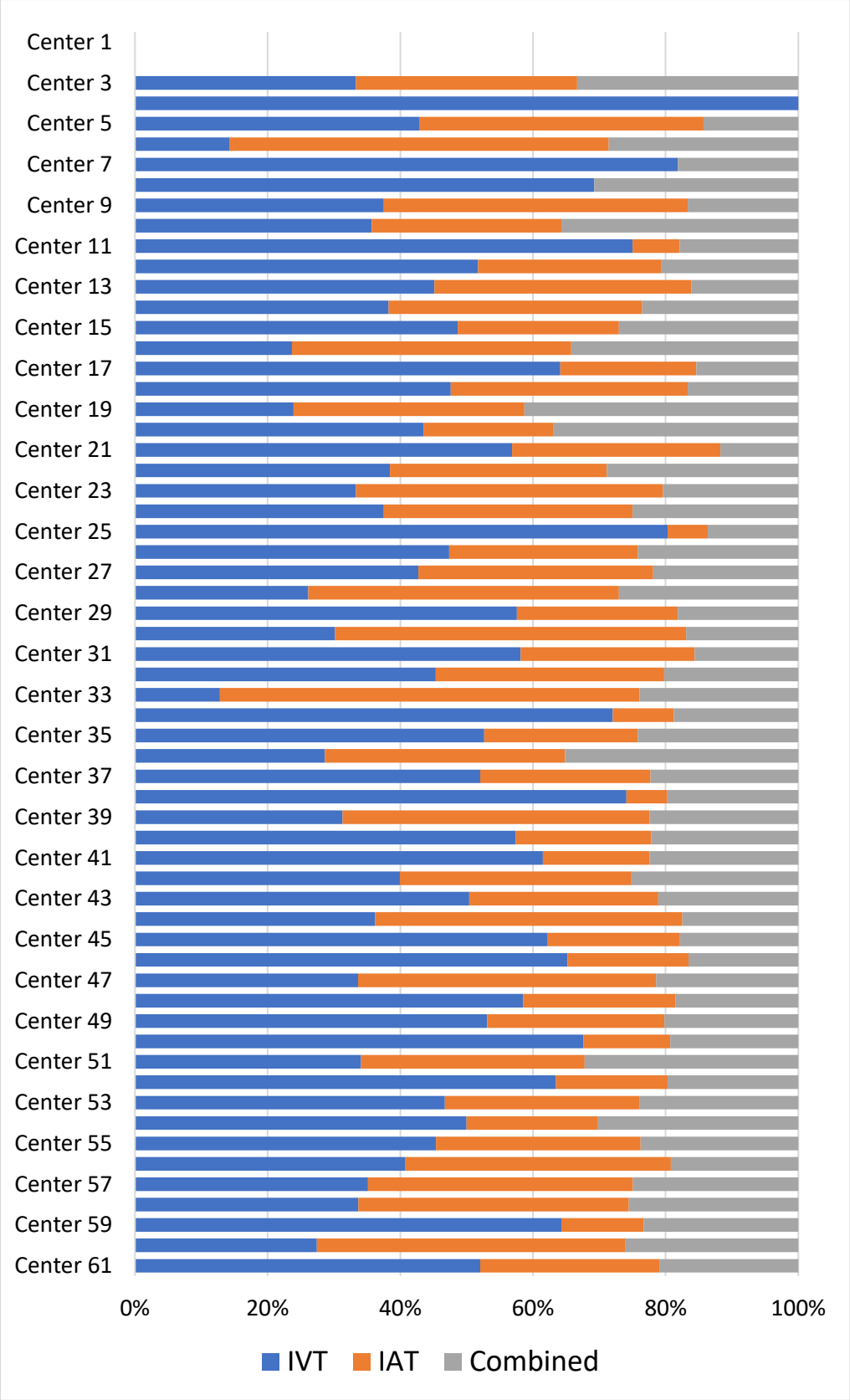
	Mean (SD)					
	Overall	2012-2013	2014-2015	2016-2017	2018-2019	2020-2021
Blood Urea Nitrogen (mg/dL)	17.7 (9.2)	17.5 (9.0)	17.5 (9.3)	17.5 (9.3)	17.6 (9.0)	18.0 (9.4)
Creatinine (mg/dL)	1.01 (1.33)	1.01 (0.92)	1.01 (1.42)	1.02 (1.11)	0.99 (0.96)	1.02 (1.77)
Hemoglobin (g/dL)	13.6 (2.1)	13.6 (2.0)	13.6 (2.0)	13.6 (2.1)	13.6 (2.1)	13.6 (2.2)
TG	127.0 (91.2)	122.8 (85.7)	122.1 (82.9)	125.8 (85.4)	129.5 (96.3)	130.5 (97.3)
LDL-cholesterol (mg/dL)	106.2 (38.6)	109.3 (37.8)	107.8 (37.9)	105.8 (38.1)	105.3 (37.4)	104.9 (40.4)
HDL-cholesterol (mg/dL)	45.7 (13.6)	44.7 (13.9)	45.1 (13.0)	45.4 (14.5)	46.0 (13.3)	46.6 (13.3)
Platelet count ($x10^3$µL)	230.1 (78.3)	231.4 (82.5)	226.4 (87.5)	231.1 (77.2)	232.3 (74.8)	229.1 (74.1)
Total cholesterol (mg/dL)	172.3 (46.1)	175.2 (43.6)	173.8 (44.0)	171.7 (45.5)	170.4 (45.3)	172.0 (49.5)
Initial Glucose (mg/dL)	145.7 (64.4)	144.6 (63.7)	144.9 (65.9)	145.2 (63.5)	146.2 (64.6)	146.5 (64.3)
Fasting Glucose (mg/dL)	123.8 (50.5)	125.3 (53.3)	121.1 (50.0)	121.7 (48.5)	124.7 (50.6)	125.1 (50.6)
Hb A1c (%)	6.35 (1.86)	6.44 (1.43)	6.36 (1.61)	6.31 (1.87)	6.31 (2.54)	6.35 (1.43)

Based on 128,609 ischemic stroke cases



Based on 124, 247 ischemic stroke cases

Figure 9. Proportion of acute revascularization treatment for each stroke center (Anonymized)



IVT: Intravenous thrombolysis; IAT: Intra-arterial thrombolysis

Figure 10. Discrepancy in revascularization modalities among centers

Table 15. Regional disparity of time interval between symptom onset and hospital arrival

	Percent of patients, %									
	Seoul	Gyeonggi & Incheon	Gyeongnam & Busan	Chungnam & Daejeon	Gyeongbuk & Daegu	Gangwon	Jeonbuk	Gwangju & Jeonnam	Chungbuk	Jeju
≤3 hour	22.1	24.4	27.6	28.2	23.3	24.9	19.7	27.1	28.9	28.5
3-6 hour	9.1	10.0	13.0	10.0	11.1	11.5	10.1	16.8	10.5	11.6
6-12 hour	13.3	13.1	14.8	13.8	13.2	13.1	13.3	15.3	14.2	14.7
12-24 hour	17.4	17.5	17.1	17.0	18.5	18.1	19.1	14.8	17.6	17.1
24-36 hour	7.9	7.2	6.7	8.0	7.3	7.0	9.0	8.5	6.3	7.0
36-48 hour	5.8	5.1	4.0	5.0	5.4	5.4	4.0	2.8	5.5	3.9
2-7 day	20.6	18.9	14.3	16.2	17.9	17.5	17.5	12.2	15.7	16.1
>7 day	3.9	3.9	2.6	1.8	3.3	2.6	7.3	2.5	1.4	1.3

Based on 144,067 ischemic stroke cases

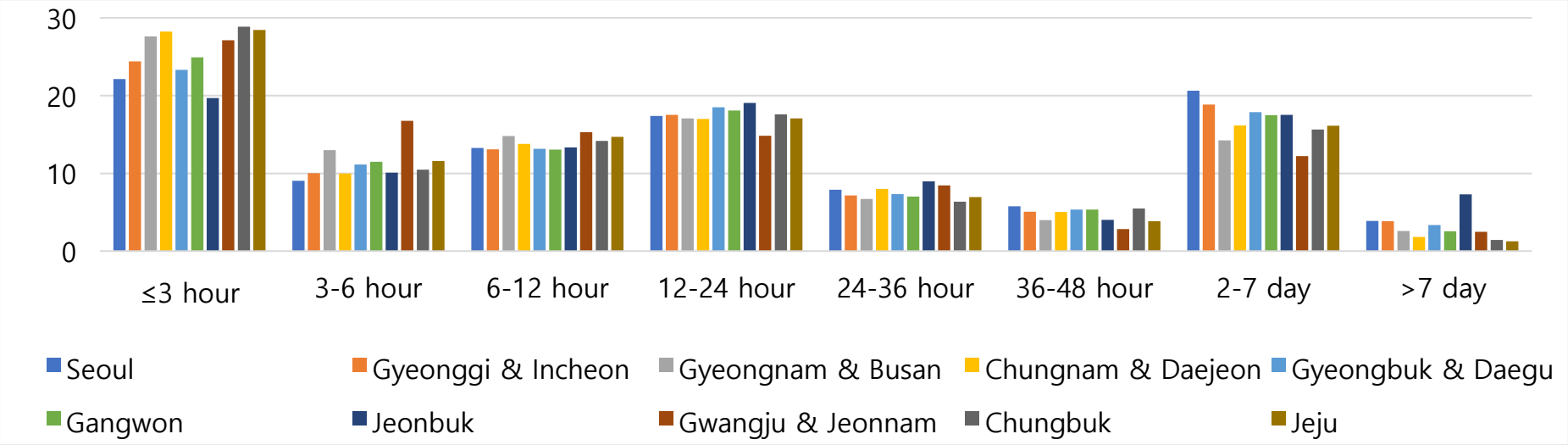


Figure 11. Regional disparity of time interval between symptom onset and hospital arrival

Table 16. Distribution and secular trends of mRS score at discharge

mRS	Number of patients (%)										
	Overall	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
0	18858 (14.9)	1261 (15.3)	1295 (14.1)	1515 (16.2)	1730 (16.1)	1993 (17.7)	1893 (15.3)	2067 (14.6)	2280 (14.4)	2025 (13.3)	2799 (13.9)
1	33178 (26.2)	2054 (25.0)	2211 (24.0)	2292 (24.6)	2669 (24.9)	2760 (24.5)	3162 (25.6)	3632 (25.6)	4545 (28.6)	4179 (27.4)	5674 (28.2)
2	24420 (19.3)	1554 (18.9)	1991 (21.6)	1862 (20.0)	2309 (21.6)	2215 (19.7)	2478 (20.1)	2652 (18.7)	2848 (17.9)	2909 (19.0)	3602 (17.9)
3	19185 (15.2)	1292 (15.7)	1482 (16.1)	1372 (14.7)	1609 (15.0)	1625 (14.4)	1814 (14.7)	2306 (16.3)	2377 (15.0)	2301 (15.1)	3007 (15.0)
4	17302 (13.7)	1189 (14.5)	1284 (13.9)	1332 (14.3)	1410 (13.2)	1527 (13.6)	1629 (13.2)	1818 (12.8)	2145 (13.5)	2101 (13.8)	2867 (14.3)
5	11238 (8.9)	813 (9.9)	853 (9.3)	849 (9.1)	837 (7.8)	932 (8.3)	1127 (9.1)	1351 (9.5)	1345 (8.5)	1374 (9.0)	1757 (8.7)
6	2369 (1.9)	59 (0.7)	101 (1.1)	110 (1.2)	149 (1.4)	216 (1.9)	247 (2.0)	351 (2.5)	344 (2.2)	388 (2.5)	404 (2.0)

Based on 126,550 ischemic stroke cases.

mRS: modified Rankin Scale

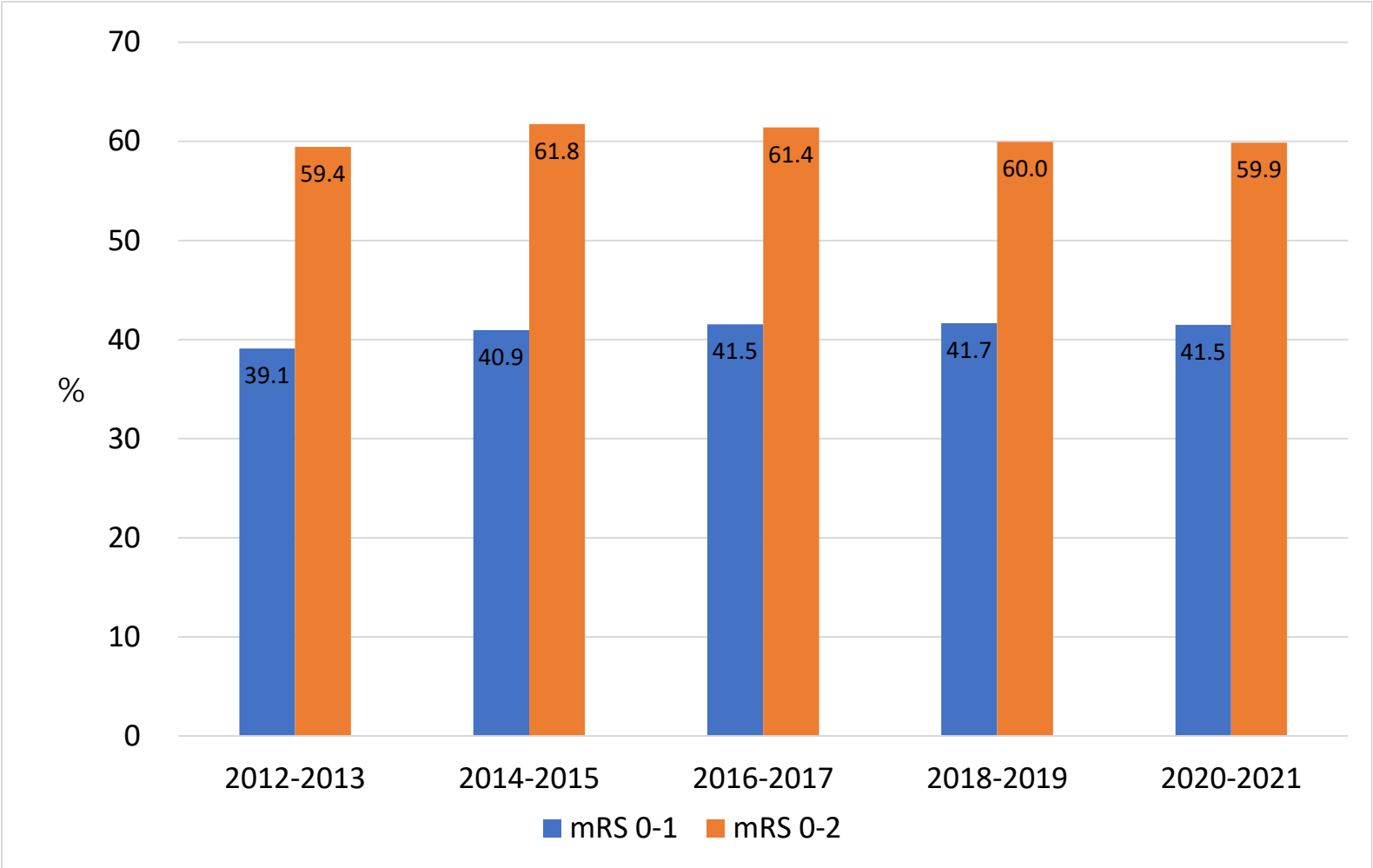


Figure 12. Secular trends of proportion in mRS 0-1 and 0-2 score at discharge

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Korean Stroke Registry
한국뇌졸중등록사업

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Korean Stroke Society

Korea Disease Control and Prevention Agency
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